Neotech Project Implementation Plan

Start Date: 24 April 2026 | Working Days: Monday to Saturday | Total Duration: 10 Days

Project Architecture Overview

Dual Layer System

Layer	Technology	Modules
Management Layer	PostgreSQL	Admin, Users, Vendors, Patients, Orders, Test Master, Samples, Procurement, Reports Management
Genetic Core Layer	MongoDB	Genetic Variant Database, Gene Reference Pages, Gene Descriptions, Patient Additional Data, All Report Templates

Full Timeline With Exact Dates

Week 1: Foundation & Core Systems

Date	Day	Module	Milestone	Status
24 April	Thursday	Orders Final	Orders, Samples flow fully operational	Planned
25 April	Friday	Admin Base	Users, Vendors, Roles, Permissions	Planned
26 April	Saturday	Patient System	Patient Additional data sync layer	Planned
27 April	Sunday	OFF	Rest Day	Rest
28 April	Monday	Report Engine	Universal report engine, PDF pipeline, MongoDB integration completed	Planned
29 April	Tuesday	Procurement	Basic stock management implemented	Planned

Report Implementation (4 Days Exactly)

Date	Day	Batch	Reports Completed	Daily Total	Status
30 April	Wednesday	Priority 1	Cardiometabolic Test, My Wellness, Woman's Health Test	**3 Reports**	Planned
1 May	Thursday	Priority 2	Men's Health, My Skin, My Immunity, Eyes Health, Kidney Health	**5 Reports**	Planned

Date	Day	Batch	Reports Completed	Daily Total	Status
2 May	Friday	Standard	Autoimmune, Sleep Test	**2 Reports**	Planned
3 May	Saturday	Special PGx	Clopidogrel, Statin, Warfarin, Hypertension-PGx	**4 Reports**	Planned

MILESTONE 3 May 2026: ALL 13 REPORTS IMPLEMENTED

> Note: Report generation will take time for each patient due to genetic variant analysis, this is expected behaviour.

Final Stabilization Week

Date	Day	Task	Status
4 May	Sunday	**OFF**	Rest
5 May	Monday	End to End testing, cross module integration	Planned
6 May	Tuesday	Bug fixes, accuracy validation, performance optimizations	Planned
7 May	Wednesday	Working days calendar implementation, final UAT preparation	Planned
8 May	Thursday	Production deployment, go-live	Planned

Final Deadlines

Item	Delivery Date
Management Layer Complete	29 April
All 13 Reports Working	3 May
Full System Ready	7 May
Production Go-Live	8 May

Technical Implementation Guidelines

1. **Next.js Project Structure**
    - Management layer components will use Drizzle ORM with PostgreSQL
    - Genetic core modules will use Mongoose with MongoDB inside same Next.js project
    - Unified API layer with proper boundary separation between both databases
    - No external microservices required, all logic will live within single Next.js application
2. **Data Flow Pattern**

```
Order Created → Sample Received → Genetic Data Upload → Variant Processing
→ Report Generation → Final Report Delivery
```

1. **Critical Constraints**
    - All dates calculations will follow Mon-Sat working days only
    - Report generation jobs will be queued as background processes
    - Genetic data will never be stored in PostgreSQL
    - Patient identifiers will be synchronized across both databases

Test Types Inventory

Total 13 Report Types:

1. Cardiometabolic Test
2. My Wellness
3. My Skin
4. My Immunity Test
5. Woman's Health Test
6. Men's Health Test
7. Eyes Health Test
8. Kidney Health Test
9. Autoimmune
10. Sleep Test
11. Clopidogrel Sensitivity (Special)
12. Statin Report (Special)
13. Warfarin Report (Special)
14. Hypertension-PGx (Special)

Dependencies & Risks

Risk	Mitigation
Report generation performance	Implement incremental generation + caching
Database synchronization	Event based triggers with retry mechanism
Data accuracy	Double validation layer for all genetic mappings

Risk	Mitigation
Legacy data migration	Phased migration with rollback points

*Last Updated: 23 April 2026